Cycle Gravity Spectra and a Uniform Counterexample Family for Graffiti 197

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Abstract. Roucairol and Cazenave exhibited the cycle on 17 vertices as a counterexample to Graffiti Conjecture 197 and conjectured that every cycle of order $4r+1 \geq 17$ is likewise a counterexample, under the usual spectral-range convention. We establish a stronger uniform conclusion: every cycle C_n with $n \geq 15$ violates the proposed inequality. The proof decomposes a shifted reciprocal-distance matrix into an explicit positive-semidefinite sum of cyclic interval Gram matrices, yielding the uniform bound $\lambda_{\min}(R_n) \geq -3/2$. A harmonic-number comparison completes both parity cases without asymptotics. Exact rational interval certificates verify that the threshold is sharp within the class of cycles: $C_3, \ldots, C_{14}$ do not violate the inequality. We supply an independently executable verifier, including exact finite checks and independent Fourier spectral tests. Attribution and historical priority are separated from mathematical correctness pending independent review.

Keywords. spectral graph theory; Graffiti conjectures; reciprocal-distance matrix; circulant matrix; positive semidefinite decomposition; certified computation.

1 The problem and its conventions

The Graffiti programme proposed many inequalities connecting graph invariants; their formulation and automated falsification have a long history [1, 2, 3]. Roucairol and Cazenave [5, §5.1] reported C_{17} as a counterexample to Graffiti 197, and also checked C_{21} and C_{25}. They conjectured that every C_{4r+1} of order at least 17 is a counterexample. Their discussion emphasizes a genuine terminological ambiguity: *range* can mean a numerical span or the number of distinct values. Our claims use *only* the numerical span, as in their counterexample.

Let C_n ($n \geq 3$) have vertex set $\mathbb{Z}/n\mathbb{Z}$, ordinary graph distance $d(u,v) = \min(|u-v|, n-|u-v|)$, and adjacency matrix A_n. If $\lambda_1(A_n) \geq \cdots \geq \lambda_n(A_n)$, write $\lambda_{n-1}(A_n)$ for its second-smallest eigenvalue. Following the gravity convention reproduced in [5, §5.2], every cycle vertex has degree two and hence

$$(\mathrm{Gr}_n)_{uv} = \begin{cases} 0 & u = v, \\ \dfrac{4}{(n-1)d(u,v)} & u \neq v. \end{cases} \tag{1}$$

Let R_n denote the reciprocal-distance matrix, so $\mathrm{Gr}_n = 4R_n/(n-1)$. Define $\mathrm{spr}(M) = \lambda_{\max}(M) - \lambda_{\min}(M)$ for symmetric M. In this convention Graffiti 197 asks for

$$-\lambda_{n-1}(A_n) \leq \mathrm{spr}(\mathrm{Gr}_n). \tag{2}$$

We show that its strict reverse holds on every sufficiently large cycle, with an exact threshold.

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Theorem 1 (Infinite counterexample family)**.** *For every integer $n \geq 15$,*

$$-\lambda_{n-1}(A_n) > \mathrm{spr}(\mathrm{Gr}_n). \tag{3}$$

Thus the Roucairol–Cazenave conjecture holds, with a stronger conclusion covering all residue classes of n.

The spectral estimate driving the proof concerns the Harary or reciprocal-distance matrix, an established graph invariant [4].

2 A positive-semidefinite identity

Let $m = \lfloor n/2 \rfloor$, let J be the all-ones matrix, and, for $2 \leq j \leq m$, let B_j be the $n \times n$ binary circulant matrix whose vth column is the indicator of the cyclic interval $\{v, v+1, \ldots, v+j-1\}$. Set

$$w_j = \begin{cases} \dfrac{2}{(j-1)j(j+1)} & 2 \leq j < m, \\ \dfrac{1}{m(m-1)} & j = m. \end{cases} \tag{4}$$

All these coefficients are strictly positive.

Lemma 1 (Exact Gram decomposition)**.** *For every $n \geq 4$,*

$$R_n + \frac{3}{2}I = \frac{1}{m}J + \sum_{j=2}^{m} w_j B_j B_j^{\mathsf{T}}. \tag{5}$$

In particular $\lambda_{\min}(R_n) \geq -3/2$.

Proof. Two cyclic intervals of length $j \leq m$ at distance $d \leq m$ share precisely $(j-d)_+ = \max(j-d, 0)$ vertices. Hence the (u,v) entry of $B_j B_j^{\mathsf{T}}$ equals $(j - d(u,v))_+$. For $1 \leq d \leq m$, the discrete convex interpolation of $f_d = 1/d - 1/m$ satisfies

$$\frac{1}{d} - \frac{1}{m} = \sum_{j=d+1}^{m} w_j (j-d),$$

with the coefficients in (4): for $j < m$ they are $1/(j-1) - 2/j + 1/(j+1)$, and the terminal coefficient is $1/(m-1) - 1/m$. Thus the off-diagonal entries in (5) equal $1/d$. On the diagonal,

$$\frac{1}{m} + \sum_{j=2}^{m} j w_j = \frac{1}{m} + \sum_{j=2}^{m-1} \left(\frac{1}{j-1} - \frac{1}{j+1} \right) + \frac{1}{m-1} = \frac{3}{2},$$

including the case $m = 2$ (empty middle sum). This proves the matrix identity. Both J and each $B_j B_j^{\mathsf{T}}$ are positive semidefinite, giving the eigenvalue bound. □

The identity is finite-dimensional and exact; its verification requires no numerical eigensolver. Let $H_m = \sum_{j=1}^{m} j^{-1}$. As R_n is nonnegative with constant row sums, its largest eigenvalue is that row sum. In particular,

$$\lambda_{\max}(R_n) = \begin{cases} 2H_m & n = 2m+1, \\ 2H_{m-1} + 1/m & n = 2m. \end{cases} \tag{6}$$

Combining (1), Lemma 1 and (6) yields

$$\operatorname{spr}(\mathrm{Gr}_n) \le \frac{4}{n-1} \left(\lambda_{\max}(R_n) + \frac{3}{2} \right). \tag{7}$$

3 Proof of the universal statement

For cycles the adjacency spectrum is exactly

$$\{ 2\cos(2\pi k/n) : k = 0, 1, \ldots, n-1 \}.$$

Ordering these eigenvalues, with multiplicity, gives

$$-\lambda_{n-1}(A_n) = \begin{cases} 2\cos(\pi/(2m+1)) & n = 2m+1, \\ 2\cos(\pi/m) & n = 2m. \end{cases} \tag{8}$$

Proof of Theorem 1. For odd $n = 2m+1$ ($m \ge 7$), (7) bounds the gravity spread by

$$q_o(m) = \frac{4H_m + 3}{m}.$$

This is strictly decreasing: $q_o(m) > q_o(m+1)$ follows from $4H_m + 3 > 4m/(m+1)$. Meanwhile $2\cos(\pi/(2m+1))$ is increasing in m. It therefore suffices to check $m = 7$. Using $H_7 = 363/140$, $\cos x \ge 1 - x^2/2$, and the rational inequality $\pi < 22/7$, we get

$$2\cos(\pi/15) - q_o(7) > \left[2 - \left(\frac{22}{105} \right)^2 \right] - \frac{468}{245} = \frac{506}{11025} > 0.$$

This proves the odd case.

For even $n = 2m$ ($m \ge 8$), define

$$S_m = 2H_{m-1} + \frac{1}{m} + \frac{3}{2}, \qquad q_e(m) = \frac{4S_m}{2m-1}.$$

Now $S_{m+1} - S_m = 1/m + 1/(m+1)$; thus $(2m-1)(S_{m+1} - S_m) < 4 < 2S_m$ for $m \ge 2$, proving $q_e(m) > q_e(m+1)$. The quantity $2\cos(\pi/m)$ increases. At $m = 8$,

$$2\cos(\pi/8) - q_e(8) > \left[2 - \left(\frac{11}{28} \right)^2 \right] - \frac{1907}{1050} = \frac{1733}{58800} > 0.$$

Both estimates compare the left-hand side of (3) to an *upper bound* on its right-hand side. The strict inequality follows for all $n \ge 15$. □

Corollary 1. *For every $r \ge 4$, the cycle C_{4r+1} refutes Graffiti 197 in the numerical-span convention, as conjectured in [5].*

4 Sharpness and certificate design

To check the finite complement without trusting floating-point eigenvalues, use a Fourier eigenvalue of R_n at frequency $k = \lfloor n/2 \rfloor$:

$$\eta_k = 2 \sum_{d=1}^{\lfloor (n-1)/2 \rfloor} \frac{\cos(2\pi k d/n)}{d} + \mathbf{1}_{2 \mid n} \frac{(-1)^k}{n/2}. \tag{9}$$

We have $\lambda_{\min}(R_n) \le \eta_k$, regardless of whether this mode is globally minimal. It follows that

$$-\lambda_{n-1}(A_n) - \operatorname{spr}(\mathrm{Gr}_n) \le 2\cos\theta_n - \frac{4}{n-1}(s_n - \eta_k), \tag{10}$$

where $s_n = \lambda_{\max}(R_n)$ and $\theta_n = \pi/n$ for odd n, $2\pi/n$ for even n.

The accompanying Python verifier encloses π by exact rational alternating-series estimates from Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. Every cosine is enclosed by exact rational evaluations of its Taylor polynomial of degree 24, with the explicit remainder bound $(22/7)^{25}/25!$. It then checks that the *upper* bound on the right-hand side of (10) is strictly negative for each $n = 3, \ldots, 14$. All acceptance-path comparisons are between exact Python rational numbers. Thus none of these twelve smaller cycles refutes (2).

Theorem 2 (Sharp cycle threshold, computer-assisted finite part)**.** *Among simple cycle graphs C_n ($n \ge 3$), violation of (2) occurs if and only if $n \ge 15$. The positive direction has the analytic proof above; the finite converse is certified by the rational interval algorithm just described.*

5 Formal certification: progressive proof record

A reproducible Lean 4/mathlib v4.24.0 implementation now certifies several of the exact algebraic and combinatorial identities needed for Lemma 1. The formal sources

and the resolved dependency manifest are retained at the public Newton Astro Labs research repository. We distinguish each kernel-checked lemma from the *not yet kernel-certified* complete spectral graph theorem.

Stage 1: positive Gram mechanism. The module `NewtonConjectures.lean` certifies the two strict rational comparison margins 506/11025 and 1733/58800, the reciprocal second-difference identity, the positivity of the interior weights, and nonnegativity of abstract weighted Gram quadratic forms. These are algebraic and rational statements; they do not alone certify the cyclic spectral conclusion.

Stage 2: universal reciprocal-distance coefficients. The module `CycleKernel.lean` proves the finite telescoping identity for every function $f : \mathbb{N} \to \mathbb{Q}$ and specializes it to $f(j) = 1/j$. For $1 \le d < m$ its exact certificate is

$$\frac{1}{d} = \frac{1}{m} + \sum_{j=d+1}^{m-1} (j-d)\frac{2}{(j-1)j(j+1)} + \frac{m-d}{m(m-1)}. \tag{11}$$

Stage 3: genuine cyclic matrix products. The modules `CyclicIntersection.lean`, `CyclicZMod.lean`, and `CyclicGram.lean` define the cyclic incidence matrix over $\mathbb{Z}/n\mathbb{Z}$, using $B_j(u,k) = 1$ exactly when the residue $k-u$ lies in $\{0,\dots,j-1\}$. For *all* vertices u, v and $2j \le n$, the Lean kernel proves

$$(B_j B_j^{\mathsf{T}})_{uv} = \max\{j - d(u,v), 0\}. \tag{12}$$

The formal proof handles cyclic wraparound, disjoint intervals, and even-order antipodes, and evaluates an actual `Matrix.mul`, rather than merely asserting a counting analogy.

Stage 4: diagonal normalization. The module `GramDiagonal.lean` proves for every $m \ge 2$ that

$$\frac{1}{m} + \sum_{j=2}^{m-1} \frac{2}{(j-1)(j+1)} + \frac{1}{m-1} = \frac{3}{2}. \tag{13}$$

It additionally certifies the corresponding diagonal *of the actual weighted incidence Gram matrices* for every $n \ge 4$ and every vertex, with $m = \lfloor n/2 \rfloor$.

Stage 5: universal off-diagonal sum. The new module `OffDiagonal.lean` proves for every $m \ge 2$ and every $1 \le d \le m$ that

$$\frac{1}{m} + \sum_{j=2}^{m} w_j (j-d)_+ = \frac{1}{d}, \tag{14}$$

where $(x)_+ = \max(x, 0)$ and the w_j are precisely those of (4). Importantly, the formal statement uses natural truncated subtraction, so the $d = m$ endpoint is included and no sign convention is silently assumed.

Independent replay and kernel gate. The Stage 5 GitHub Actions run 39131413230 completed all 7,358 Lean build jobs. An independent axiom audit accepted 100 declarations using only the foundational allowlist `propext`, `Classical.choice`, and `Quot.sound`; the proof-placeholder check passed. In a separate exact `fractions.Fraction` replay, all 7,259 pairs (m, d) with $2 \le m \le 120$ and $1 \le d \le m$ passed (14). No floating-point acceptance criterion was used for these claims. The complete Lean proof ledger separates verified and outstanding obligations.

Precise boundary of formal certification. Equations (12), (13), and (14), taken together, supply entrywise components of (5). However, the single all-entry matrix theorem and the subsequent cycle-specific spectral bound must still be assembled and checked *inside Lean*. In particular, the ordering of the cycle adjacency eigenvalues, the two uniform harmonic comparisons, and the exact finite converse $C_3, \dots, C_{14}$ have not all been promoted to a complete Lean proof. The analytical theorem and the finite exact certificates are presented independently of this partial formalization.

6 Reproducibility and claim boundaries

The paper's proof is self-contained apart from classical elementary facts ($\pi < 22/7$, $\cos x \ge 1 - x^2/2$, and the cycle adjacency spectrum). The accompanying `verify.py` provides four checks: rational matrix-identity entries for orders 4 to 40, rational bounds for all small orders 3 to 14, rational evaluation of both boundary inequalities, and an independent floating-point Fourier cross-check through order 160. The numerical cross-check is deliberately *not* an ingredient of the theorem. Proof-critical component lemmas have now been independently kernel-checked in Lean 4/mathlib as documented in Section 5; the complete graph-theoretic theorem has not been formally certified.

Graffiti 197 itself was already refuted by Roucairol and Cazenave [5]; its resolution is not claimed as a new discovery. The contribution under review is the uniform analytic infinite-family theorem, the extended threshold, and the exact Gram decomposition. The targeted literature audit did not identify an earlier proof of this specific strengthening, but that negative search result is not a priority certificate. Mathematical validity, originality, and suitability for publication remain open to independent assessment.

References

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